

Suman Datta

professor

基本信息

姓名: Suman Datta

所属院校: 圣母玛利亚大学

邮箱: sdatta@nd.edu

地区: United States

年龄范围: 50岁以上

海外经历: 有

企业经历: 有

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

集成电路设计与集成系统

微电子科学与工程

研究领域

低功耗电路与能量管理

微电子器件与制造

存储器设计与架构

数字集成电路设计

纳米电子与先进工艺

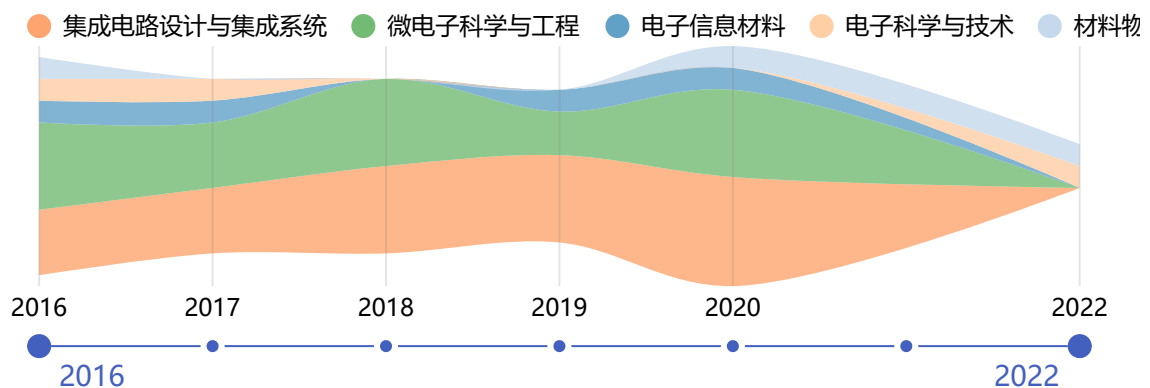
高性能计算芯片

个人简介

Suman Datta is the Stinson Professor of Nanotechnology at the University of Notre Dame and the Director of ASCENT. He holds a B.E. from IIT Kanpur (1995) and a Ph.D. from the University of Cincinnati (1999). He worked at Intel from 1999 - 2007, then joined Penn State, and was promoted to full - professor

in 2011. He has over 375 publications and 185 patents. A Fellow of NAI and IEEE, his interests include nanoelectronic device design, emerging electronics for new compute models. Office: 271 Fitzpatrick Hall, Phone: 574 - 631 - 8835, Email: sdatta@nd.edu.

近年发表的论文



Theoretical and Empirical Insight into Dopant, Mobility and Defect States in W Doped Amorphous In₂O₃ for High-Performance Enhancement Mode BEOL Transistors

2022 INTERNATIONAL ELECTRON DEVICES MEETING, IEDM

Hu, Yaoqiao; Ye, Huacheng; Aabrar, Khandker Akif; Kirtania, Sharadindu Gopal; Chakraborty, Wriddhi; Datta, Suman; Cho, Kyeongjae

被引用次数: 7

2022

Drain-Erase Scheme in Ferroelectric Field Effect Transistor-Part II: 3-D-NAND Architecture for In-Memory Computing

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wang, Panni; Shim, Wonbo; Wang, Zheng; Hur, Jae; Datta, Suman; Khan, Asif Islam; Yu, Shimeng

被引用次数: 35

2020

Drain-Erase Scheme in Ferroelectric Field-Effect Transistor-Part I: Device Characterization

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wang, Panni; Wang, Zheng; Shim, Wonbo; Hur, Jae; Datta, Suman; Khan, Asif Islam; Yu, Shimeng

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2020

Investigating Ferroelectric Minor Loop Dynamics and History Effect-Part II: Physical Modeling and Impact on Neural Network Training

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wang, Panni; Wang, Zheng; Sun, Xiaoyu; Hur, Jae; Datta, Suman; Islam Khan, Asif; Yu, Shimeng

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Investigating Ferroelectric Minor Loop Dynamics and History Effect-Part I: Device Characterization

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Wang, Panni; Wang, Zheng; Sun, Xiaoyu; Hur, Jae; Datta, Suman; Khan, Asif Islam; Yu, Shimeng

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2020

FerroElectronics for Edge Intelligence

IEEE MICRO

Keshavarzi, Ali; Ni, Kai; van den Hoek, Wilbert; Datta, Suman; Raychowdhury, Arijit

被引用次数: 61

2020

Mismatch of Ferroelectric Film on Negative Capacitance FETs Performance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Liang, Yuhua; Zhu, Zhangming; Li, Xueqing; Gupta, Sumeet Kumar; Datta, Suman; Narayanan, Vijaykrishnan

被引用次数: 25

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The future of ferroelectric field-effect transistor technology

NATURE ELECTRONICS

Khan, Asif Islam; Keshavarzi, Ali; Datta, Suman

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2020

An Ultra-Dense 2FeFET TCAM Design Based on a Multi-Domain FeFET Model

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yin, Xunzhao; Ni, Kai; Reis, Dayane; Datta, Suman; Niemier, Michael; Hu, Xiaobo Sharon

被引用次数: 105

2019

SRAMs and DRAMs With Separate Read-Write Ports Augmented by Phase Transition Materials

IEEE TRANSACTIONS ON ELECTRON DEVICES

Shen, Zhesheng; Srinivasa, Srivatsa; Aziz, Ahmedullah; Datta, Suman; Narayanan, Vijaykrishnan; Gupta, Sumeet Kumar

被引用次数: 7

2019

Utilization of Negative-Capacitance FETs to Boost Analog Circuit Performances

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Liang, Yuhua; Zhu, Zhangming; Li, Xueqing; Gupta, Sumeet Kumar; Datta, Suman; Narayanan, Vijaykrishnan

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2019

Back-End-of-Line Compatible Transistors for Monolithic 3-D Integration

IEEE MICRO

Datta, Suman; Dutta, Sourav; Grisafe, Benjamin; Smith, Jeff; Srinivasa, Srivatsa; Ye, Huacheng

被引用次数: 102

2019

Power and Area Efficient FPGA Building Blocks Based on Ferroelectric FETs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Xiaoming; Ni, Kai; Niemier, Michael T.; Han, Yinhe; Datta, Suman; Hu, Xiaobo Sharon

被引用次数: 23

2019

Ten nanometre CMOS logic technology

NATURE ELECTRONICS

Datta, Suman

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2018

Lowering Area Overheads for FeFET-Based Energy-Efficient Nonvolatile Flip-Flops

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Li, Xueqing; George, Sumitha; Liang, Yuhua; Ma, Kaisheng; Ni, Kai; Aziz, Ahmedullah; Gupta, Sumeet Kumar; Sampson, John; Chang, Meng-Fan; Liu, Yongpan; Yang, Huazhong; Datta, Suman; Narayanan, Vijaykrishnan

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Influence of Body Effect on Sample-and-Hold Circuit Design Using Negative Capacitance FET

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Analysis of DIBL Effect and Negative Resistance Performance for NCFET Based on a Compact SPICE Model

IEEE TRANSACTIONS ON ELECTRON DEVICES

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The era of hyper-scaling in electronics

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Salahuddin, Sayeef; Ni, Kai; Datta, Suman

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Dynamic Diagnosis for Defective Reconfigurable Single-Electron Transistor Arrays

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Li, Yun-Jui; Huang, Ching-Yi; Wu, Chia-Cheng; Chen, Yung-Chih; Wang, Chun-Yao; Datta, Suman; Narayanan, Vijaykrishnan

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2017

Design of Nonvolatile SRAM with Ferroelectric FETs for Energy-Efficient Backup and Restore

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Li, Xueqing; Ma, Kaisheng; George, Sumitha; Khwa, Win-San; Sampson, John; Gupta, Sumeet; Liu, Yongpan; Chang, Meng-Fan; Datta, Suman; Narayanan, Vijaykrishnan

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2017

Advancing Nonvolatile Computing With Nonvolatile NCFET Latches and Flip-Flops

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Li, Xueqing; George, Sumitha; Ma, Kaisheng; Tsai, Wei-Yu; Aziz, Ahmedullah; Sampson, John; Gupta, Sumeet Kumar; Chang, Meng-Fan; Liu, Yongpan; Datta, Suman; Narayanan, Vijaykrishnan

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2017

Fabrication, Characterization, and Analysis of Ge/GeSn Heterojunction p-Type Tunnel Transistors

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Schulte-Braucks, Christian; Pandey, Rahul; Sajjad, Redwan Noor; Barth, Mike; Ghosh, Ram Krishna; Grisafe, Ben; Sharma, Pankaj; von den Driesch, Nils; Vohra, Anurag; Rayner, Gilbert Bruce, Jr.; Loo, Roger; Mantl, Siegfried; Buca, Dan; Yeh, Chih-Chieh; Wu, Cheng-Hsien; Tsai, Wilman; Antoniadis, Dimitri A.; Datta, Suman

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2017

Device-Circuit Analysis of Ferroelectric FETs for Low-Power Logic

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Gupta, Shreya; Steiner, Mark; Aziz, Ahmedullah; Narayanan, Vijaykrishnan; Datta, Suman; Gupta, Sumeet Kumar

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Enabling Energy-Efficient Nonvolatile Computing With Negative Capacitance FET

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2017

Single-Event Measurement and Analysis of Antimony-Based p-Channel Quantum-Well MOSFETs With High- κ Dielectric

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Barth, Michael; Kumar, Archana; Warner, Jeffrey H.; Bennett, Brian R.; Cress, Cory D.; Boos, J. Brad; Roche, Nicolas J. -H.; Raine, Melanie; Gaillardin, Marc; Paillet, Philippe; McMorrow, Dale; Saraswat, Krishna; Datta, Suman

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2017

Exploration of Low-Power High-SFDR Current-Steering D/A Converter Design Using Steep-Slope Heterojunction Tunnel FETs

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Kim, Moon Seok; Li, Xueqing; Liu, Huichu; Sampson, John; Datta, Suman; Narayanan, Vijaykrishnan

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2016

Opportunities and Challenges of Tunnel FETs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Pandey, Rahul; Mookerjee, Saurabh; Datta, Suman

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Correlated Material Enhanced SRAMs With Robust Low Power Operation

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Srinivasa, Srivatsa; Aziz, Ahmedullah; Shukla, Nikhil; Li, Xueqing; Sampson, John; Datta, Suman; Kulkarni, Jaydeep P.; Narayanan, Vijaykrishnan; Gupta, Sumeet Kumar

Revisiting the Theory of Ferroelectric Negative Capacitance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Majumdar, Kausik; Datta, Suman; Rao, Satyavolu Papa

Comparative Area and Parasitics Analysis in FinFET and Heterojunction Vertical TFET Standard Cells

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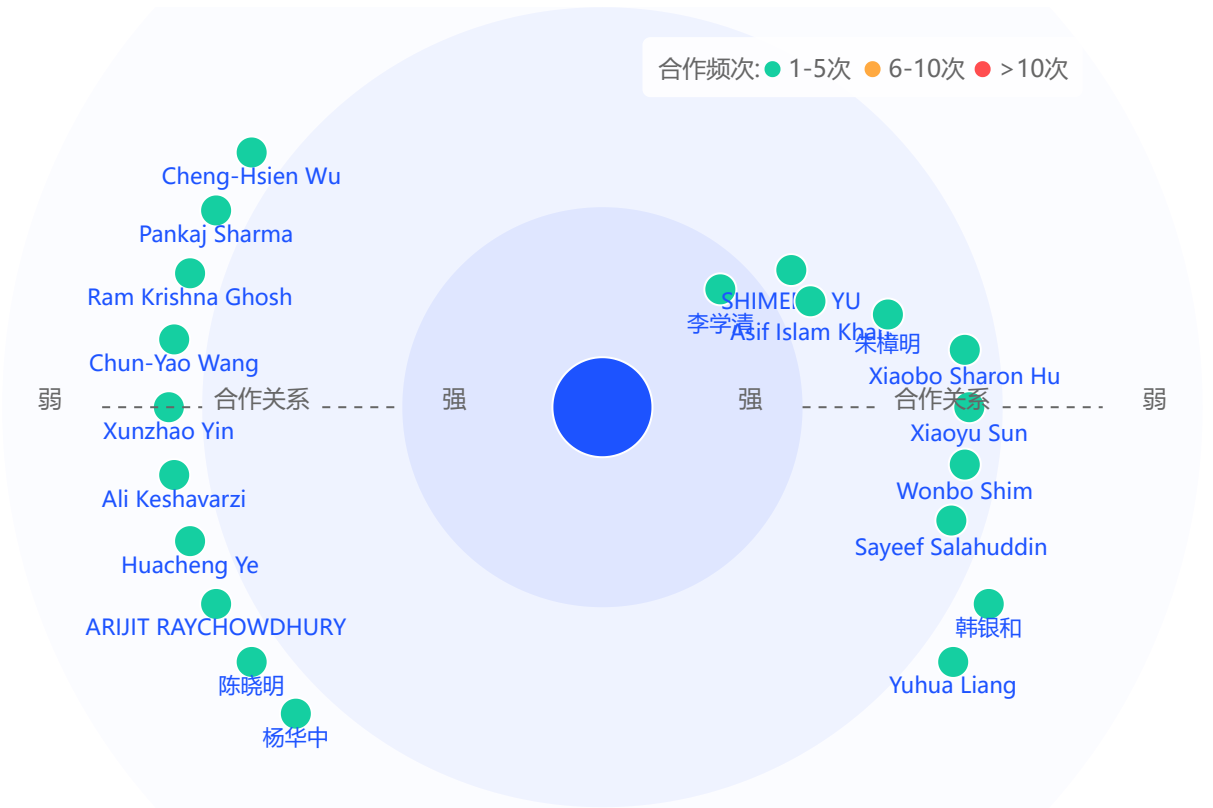
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Computing With Networks of Oscillatory Dynamical Systems

Proc.IEEE

Arijit Raychowdhury; Abhinav Parihar; Gus Henry Smith; Vijaykrishnan Narayanan; Gy?rgy Csaba; Matthew Jerry; Wolfgang Porod; Suman Datta

学术合作



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